



TASK 2 – DBMS PROJECT

GROUP 7

GROUP MEMBERS:

Doshi, Pratham Milankumar
Khanna, Rakshit
Lakhani, Vimal Shantibhai
Solanki, Neel Paresh

Neel Solanki

TASK 2.0.

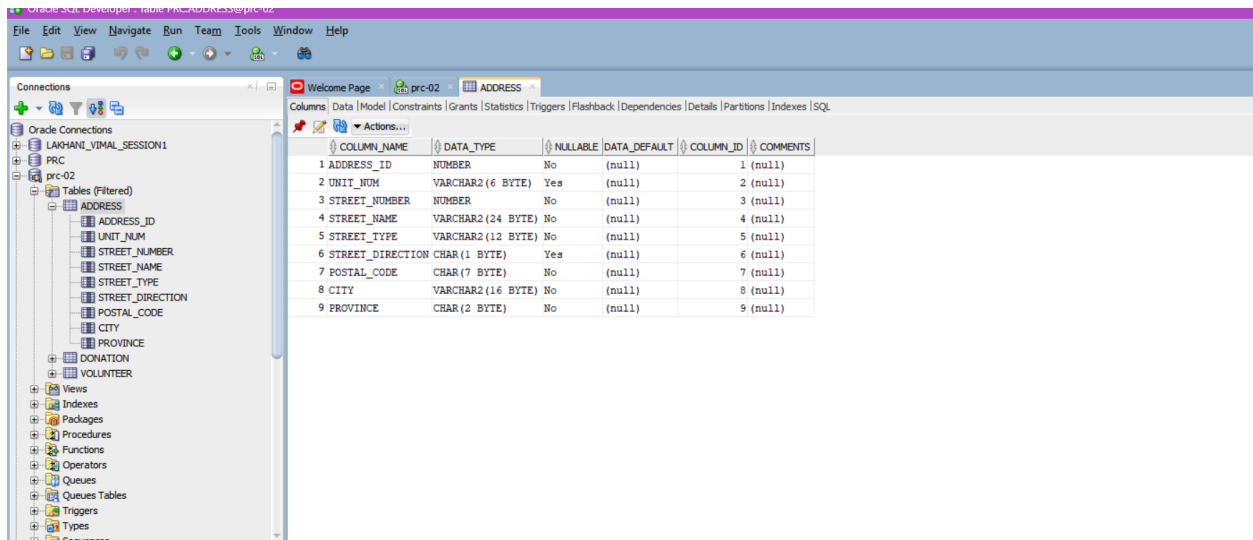
TASK DISTRIBUTION

NEEL PARESH SOLANKI	Leads the project by designing the database, preparing 3 CSVs with valid/invalid data, and building the complete Talend ETL job.
VIMAL SHANTIBHAI LAKHANI	Tests the Talend job, verifies correct insertions and rejections, and helps identify any data or mapping issues.
PRATHAM MILANKUMAR DOSHI	Assists in preparing group-specific rejected CSV outputs and reviews rejected logic by group leader.
RAKSHIT KHANNA	Compiles screenshots, prepares the final project report, and documents the ETL process and testing outcomes.

Creating Tables in Oracle SQL Developer for Task 2

1. Donation Table

```
CREATE TABLE Donation (  
    Donation_ID NUMBER PRIMARY KEY,  
    Donor_First_Name VARCHAR2(16),  
    Donor_Last_Name VARCHAR2(16),  
    Donation_Date DATE NOT NULL,  
    Doantion_Amount NUMBER(7,1) NOT NULL,  
    Address_ID NOT NULL,  
    Volunteer_ID NOT NULL,  
    CONSTRAINT fk_don_add  
    FOREIGN KEY (Address_ID) REFERENCES ADDRESS (Address_ID),  
    CONSTRAINT fk_don_vol FOREIGN KEY (Volunteer_ID)REFERENCE  
    VOLUNTEER(Volunteer_ID)
```

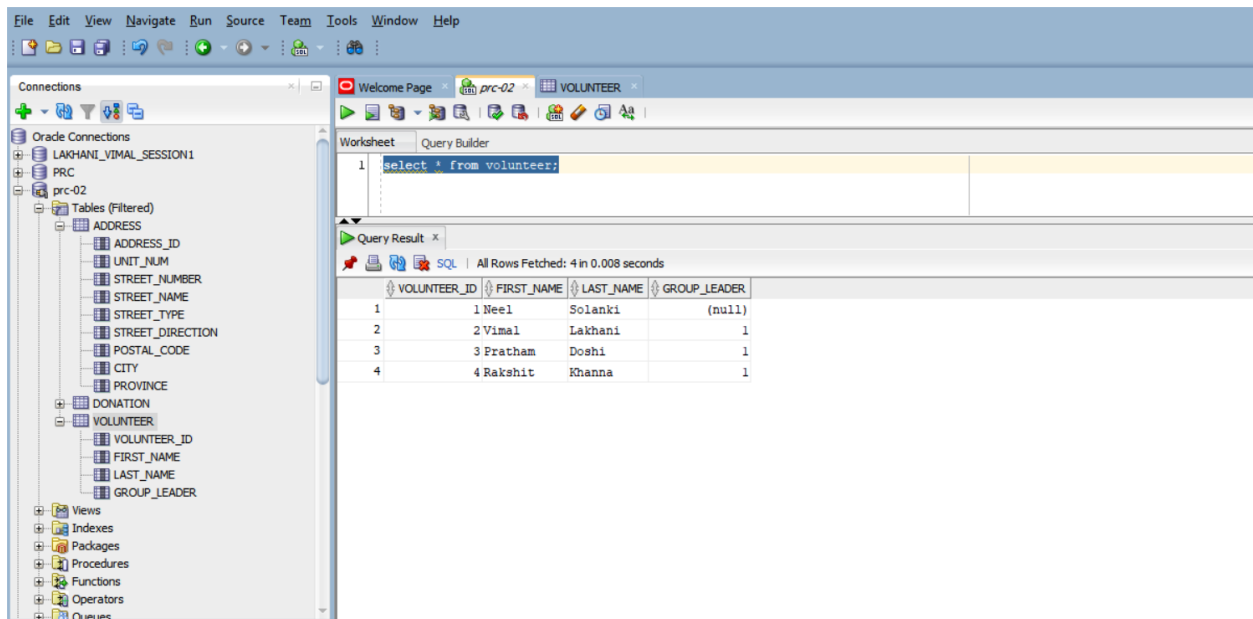
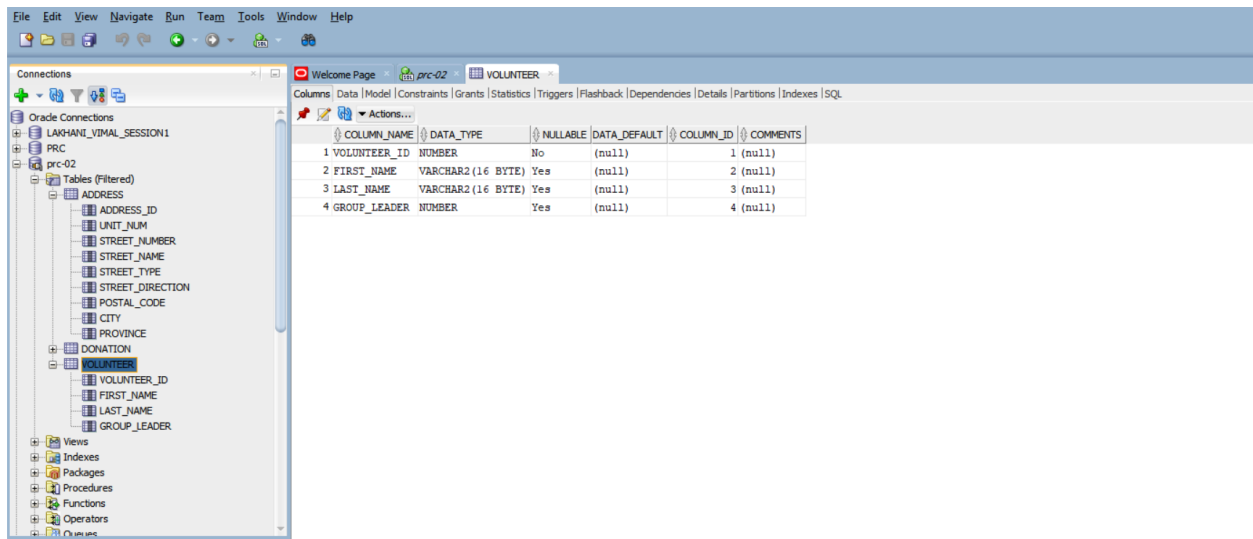


The screenshot shows the Oracle SQL Developer interface. On the left, the 'Connections' pane displays a tree view of the database schema, including tables like ADDRESS, DONATION, and VOLUNTEER. The main window shows the 'ADDRESS' table structure with the following columns:

COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
ADDRESS_ID	NUMBER	No	(null)	1	(null)
UNIT_NUM	VARCHAR2 (6 BYTE)	Yes	(null)	2	(null)
STREET_NUMBER	NUMBER	No	(null)	3	(null)
STREET_NAME	VARCHAR2 (24 BYTE)	No	(null)	4	(null)
STREET_TYPE	VARCHAR2 (12 BYTE)	No	(null)	5	(null)
STREET_DIRECTION	CHAR (1 BYTE)	Yes	(null)	6	(null)
POSTAL_CODE	CHAR (7 BYTE)	No	(null)	7	(null)
CITY	VARCHAR2 (16 BYTE)	No	(null)	8	(null)
PROVINCE	CHAR (2 BYTE)	No	(null)	9	(null)

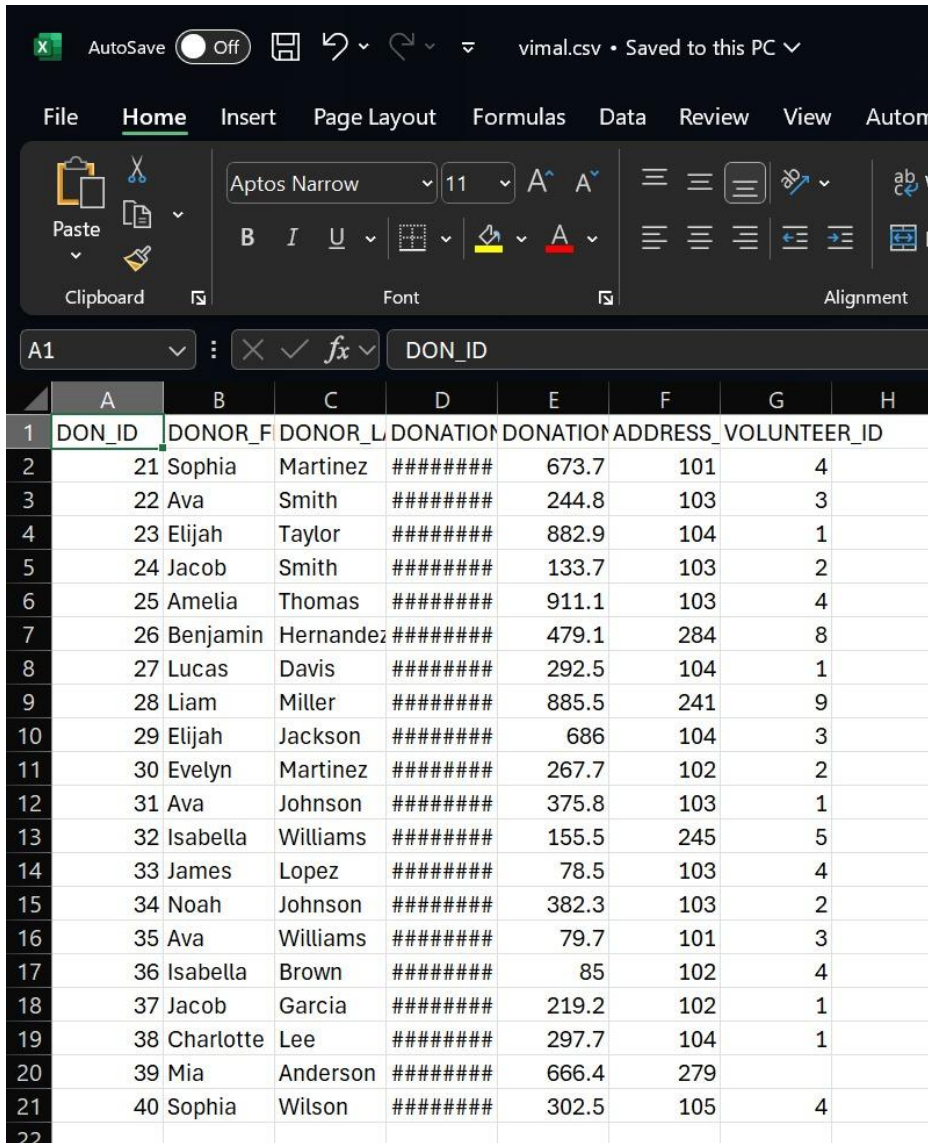
2. Volunteer Table – (for Volunteer Group Leader OR Group Member Id)

```
CREATE TABLE VOLUNTEER (  
  Volunteer_ID NUMBER PRIMARY KEY,  
  First_Name VARCHAR2(16),  
  Last_Name VARCHAR2(16),  
  Group_Leader,  
  CONSTRAINT fk_vol_lead  
  FOREIGN KEY ( Group_Leader )  
  REFERENCES VOLUNTEER(Volunteer_ID))
```



3. CSV FILES (contains VALID & INVALID data):

Vimal.csv



AutoSave Off vimal.csv • Saved to this PC

File Home Insert Page Layout Formulas Data Review View Autom

Paste Clipboard Font Alignment

A1 : X ✓ fx DON_ID

	A	B	C	D	E	F	G	H
1	DON_ID	DONOR_F	DONOR_L	DONATION	DONATION	ADDRESS	VOLUNTEER_ID	
2	21	Sophia	Martinez	#####	673.7	101	4	
3	22	Ava	Smith	#####	244.8	103	3	
4	23	Elijah	Taylor	#####	882.9	104	1	
5	24	Jacob	Smith	#####	133.7	103	2	
6	25	Amelia	Thomas	#####	911.1	103	4	
7	26	Benjamin	Hernandez	#####	479.1	284	8	
8	27	Lucas	Davis	#####	292.5	104	1	
9	28	Liam	Miller	#####	885.5	241	9	
10	29	Elijah	Jackson	#####	686	104	3	
11	30	Evelyn	Martinez	#####	267.7	102	2	
12	31	Ava	Johnson	#####	375.8	103	1	
13	32	Isabella	Williams	#####	155.5	245	5	
14	33	James	Lopez	#####	78.5	103	4	
15	34	Noah	Johnson	#####	382.3	103	2	
16	35	Ava	Williams	#####	79.7	101	3	
17	36	Isabella	Brown	#####	85	102	4	
18	37	Jacob	Garcia	#####	219.2	102	1	
19	38	Charlotte	Lee	#####	297.7	104	1	
20	39	Mia	Anderson	#####	666.4	279		
21	40	Sophia	Wilson	#####	302.5	105	4	
22								

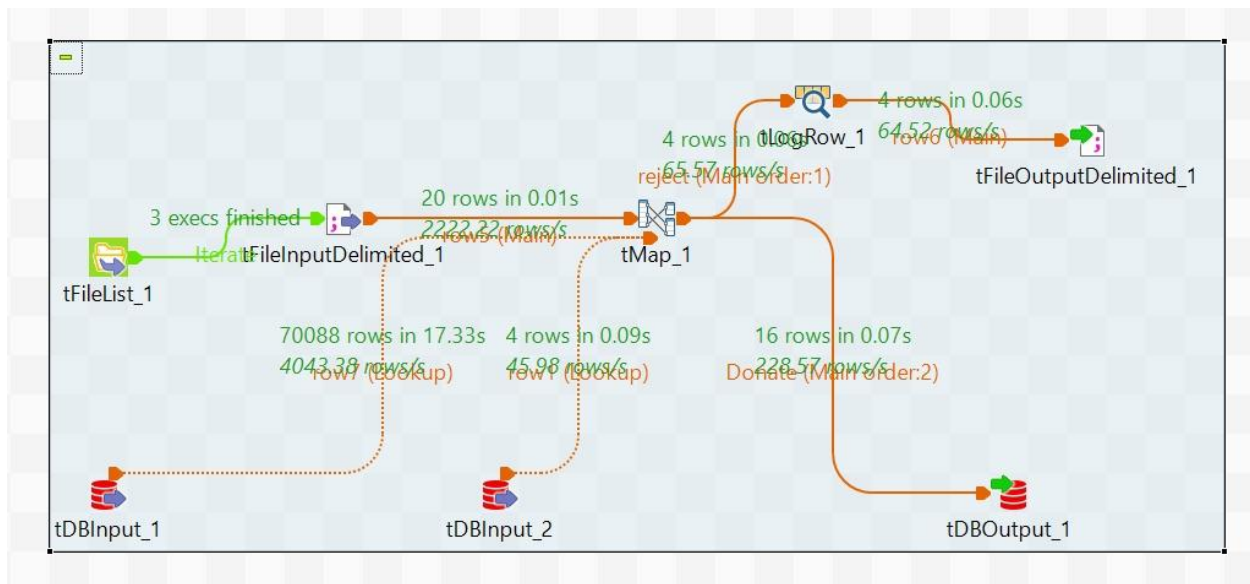
Neel.csv

A	B	C	D	E	F	G	H
1	DON_ID	DONOR_F	DONOR_LAST_NAME	DONATION_DATE	DONATION_AMOUNT	DONATION_ADDRESS	VOLUNTEER_ID
2	1	Elijah	Anderson	2023-04-28	26.9	102	4
3	2	Ethan	Hernandez	2023-08-05	822.4	229	5
4	3	Elijah	Williams	2024-07-21	932.5	105	1
5	4	Harper	Johnson	2024-04-22	176.1	102	3
6	5	Benjamin	Davis	2024-09-17	727.4	105	3
7	6	Noah	Jackson	2024-09-30	192.7	248	
8	7	Charlotte	Smith	2023-10-31	136	103	2
9	8	Elijah	Thomas	2023-10-12	560.4	101	2
10	9	Elijah	Jackson	2024-06-22	21.5	104	1
11	10	Lucas	Gonzalez	2023-05-08	853.3	101	4
12	11	Logan	Taylor	2023-10-26	424	101	1
13	12	Mia	Martinez	2023-11-24	972	104	4
14	13	Sophia	Martin	2024-06-01	855.9	102	3
15	14	Harper	Moore	2024-11-18	780.7	101	4
16	15	Amelia	Martinez	2024-10-20	734.7	104	2
17	16	Benjamin	Thomas	2023-12-13	343.6	104	3
18	17	Olivia	Garcia	2023-01-03	991.1	298	
19	18	Lucas	Miller	2024-05-18	161.4	103	2
20	19	Elijah	Smith	2023-04-30	948	101	1
21	20	Noah	Smith	2024-12-05	158.3	250	

Pratham.csv

A	B	C	D	E	F	G	H
1	DON_ID	DONOR_F	DONOR_LAST_NAME	DONATION_DATE	DONATION_AMOUNT	DONATION_ADDRESS	VOLUNTEER_ID
2	41	Charlotte	Taylor	2023-01-29	759.8	101	2
3	42	Ethan	Lee	2023-11-09	79.3	105	2
4	43	Evelyn	Anderson	2023-05-28	838.4	101	2
5	44	Mason	Thomas	2024-11-18	928.1	299	5
6	45	Benjamin	Jones	2024-09-23	932.9	102	2
7	46	Noah	Taylor	2023-06-19	117.4	105	2
8	47	Amelia	Smith	2024-04-23	821.3	102	2
9	48	Liam	Williams	2024-02-14	932.7	235	
10	49	Emma	Moore	2023-06-01	636.4	281	
11	50	Isabella	Wilson	2023-06-30	497.3	102	3
12	51	Mason	Jackson	2023-12-30	966.8	103	4
13	52	Mason	Gonzalez	2023-02-24	609.7	287	
14	53	Liam	Hernandez	2024-12-13	899.3	102	1
15	54	Evelyn	Hernandez	2024-11-10	173.5	103	2
16	55	Olivia	Martin	2023-08-15	345.3	298	
17	56	Harper	Jones	2023-12-10	500.7	105	2
18	57	Sophia	Martin	2023-08-20	368.2	223	
19	58	Elijah	Miller	2023-09-28	975.7	103	1
20	59	Logan	Smith	2023-01-08	661.2	103	3
21	60	Jacob	Martin	2023-02-05	880.2	249	

TALEND PROCESS:



Extract Phase

- **tFileList:** Iterates through all available CSV files containing donation data.
- **tFileInputDelimited:** Reads each file sequentially, processing the content line by line.

Transform Phase

- **tMap:**
 - Performs lookup operations using reference data from tDBInput_1 (ADDRESS) and tDBInput_2 (VOLUNTEER).
 - Validates foreign key relationships, specifically Address_ID and Volunteer_ID.
 - Routes records with valid foreign keys to the database output flow, while directing invalid records to a designated reject flow.

Load Phase

- **tDBOutput:** Inserts valid donation records into the DONATION table within the database.
- **tFileOutputDelimited:** Writes rejected records to a CSV file for further analysis.
- **tLogRow:** Outputs rejected records to the console for immediate visibility and debugging.

1. tMap Configuration :

Talend Open Studio for Data Integration - tMap - tMap_1

Find:

Var

row5

Column

- Don_ID
- Donor_First_name
- Donor_Last_name
- Donation_Date
- Donation_Amount
- Address_ID
- Volunteer_ID

row7

Property Value

Lookup Model Load once

Match Model Unique match

Join Model Left Outer Join

Store temp data false

Expr. key

row5.Address_ID

Column

- ADDRESS_ID
- UNIT_NUM
- STREET_NUMBER
- STREET_NAME
- STREET_TYPE
- STREET_DIRECTION
- POSTAL_CODE
- CITY
- PROVINCE

row1

Property Value

Lookup Model Load once

Match Model Unique match

Join Model Left Outer Join

Store temp data false

Schema editor - Expression editor

row5

Columns	K...	Type	N	Date Pattern (Ctrl+Spa...	Length	Precision	Default	Comment
Don_ID		BigDecimal			0	0		
Donor_First_name		String			8	0		

reject

Property Value

Catch output reject false

Catch lookup inner join reject false

Schema Type Repository

Schema Id DB (ORACLE_SERVICE_NAME)AddressOracle ...

row5.Address_ID == null || row7.Address_ID == null ||

Expression

Column

- row5.Don_ID DON_ID
- row5.Donor_First_name DONOR_FIRST_NAME
- row5.Donor_Last_name DONOR_LAST_NAME
- row5.Donation_Date DONATION_DATE
- row5.Donation_Amount DONATION_AMOUNT
- row5.Address_ID ADDRESS_ID
- row5.Volunteer_ID VOLUNTEER_ID

Donate

Property Value

Catch output reject true

Catch lookup inner join reject true

Schema Type Repository

Schema Id DB (ORACLE_SERVICE_NAME)AddressOracle ...

row1.VOLUNTEER_ID != null &&

Expression

Column

- row5.Don_ID DON_ID
- row5.Donor_First_name DONOR_FIRST_NAME
- row5.Donor_Last_name DONOR_LAST_NAME
- row5.Donation_Date DONATION_DATE
- row5.Donation_Amount DONATION_AMOUNT
- row5.Address_ID ADDRESS_ID
- row5.Volunteer_ID VOLUNTEER_ID

Schema editor - Expression editor

row5

Columns	K...	Type	N	Date Pattern (Ctrl+Spa...	Length	Precision	Default	Comment
Don_ID		BigDecimal			0	0		
Donor_First_name		String			8	0		

reject

Column

- DON_ID
- DONOR_FIRST_NAME
- DONOR_LAST_NAME
- DONATION_DATE
- DONATION_AMOUNT
- ADDRESS_ID
- VOLUNTEER_ID

Apply OK Cancel

Talend Open Studio for Data Integration - tMap - tMap_1

Find:

Var

row5

Column

- Don_ID
- Donor_First_name
- Donor_Last_name
- Donation_Date
- Donation_Amount
- Address_ID
- Volunteer_ID

row7

Property Value

Lookup Model Load once

Match Model Unique match

Join Model Left Outer Join

Store temp data false

Expr. key

row5.Address_ID

Column

- ADDRESS_ID
- UNIT_NUM
- STREET_NUMBER
- STREET_NAME
- STREET_TYPE
- STREET_DIRECTION
- POSTAL_CODE
- CITY
- PROVINCE

Schema editor - Expression editor

row5

Columns	K...	Type	N	Date Pattern (Ctrl+Spa...	Length	Precision	Default	Comment
Don_ID		BigDecimal			0	0		
Donor_First_name		String			8	0		
Donor_Last_name		String			14	0		
Donation_Date		Date		"yyyy-MM-dd"	0	0		
Donation_Amount		BigDecimal			0	0		
Address_ID		BigDecimal			0	0		
Volunteer_ID		BigDecimal			0	0		

reject

Column

- DON_ID
- DONOR_FIRST_NAME
- DONOR_LAST_NAME
- DONATION_DATE
- DONATION_AMOUNT
- ADDRESS_ID
- VOLUNTEER_ID

Apply OK Cancel

Expression Filter for invalid data (out.csv):

```
row5.Address_ID == null || row7.ADDRESS_ID == null ||
```

```
row5.Volunteer_ID == null || row1.VOLUNTEER_ID == null ||
```

```
!row5.Address_ID.equals(row7.ADDRESS_ID) ||
```

```
!row5.Volunteer_ID.equals(row1.VOLUNTEER_ID)
```

Expression Filter for Valid Data (DONATION TABLE) :

row1.VOLUNTEER_ID != null &&

row7.ADDRESS_ID != null &&

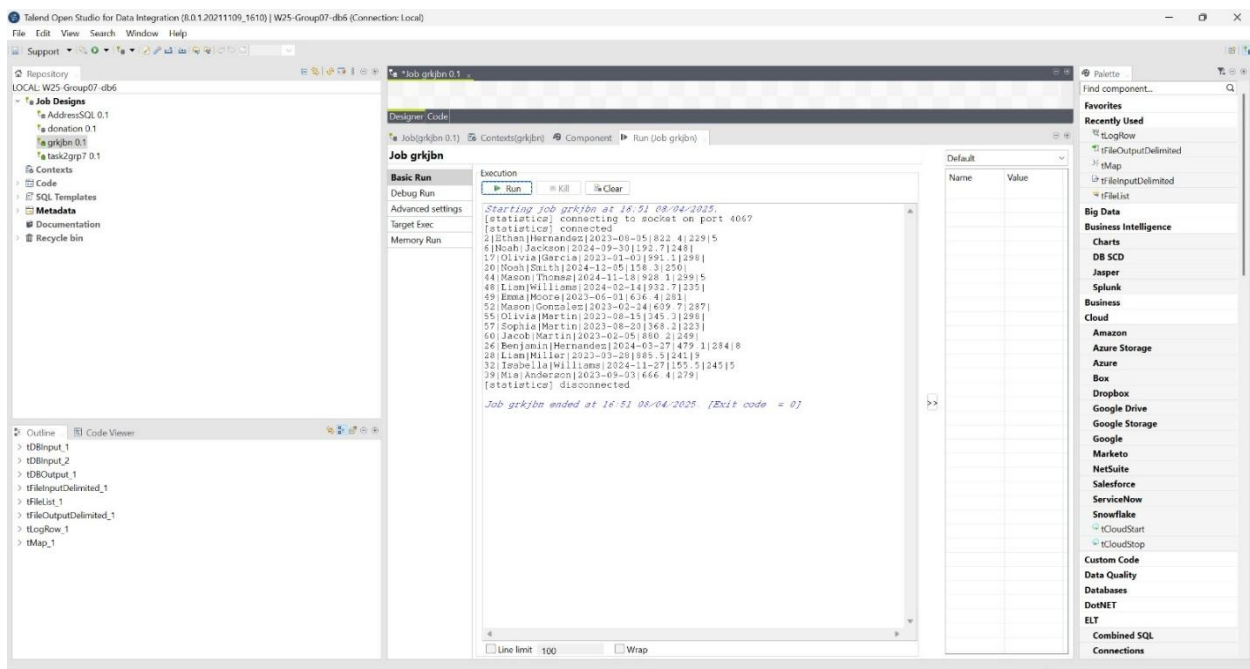
```
row5.Volunteer_ID != null &&
```

```
row5.Address_ID != null &&
```

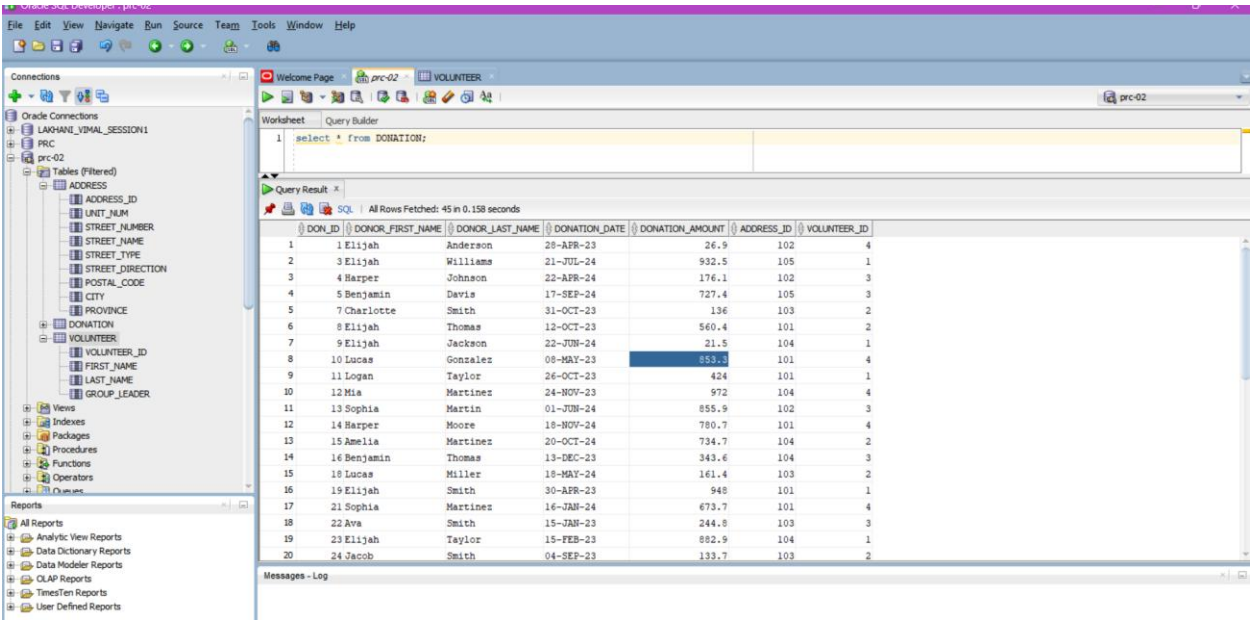
```
row1.VOLUNTEER_ID.compareTo(row5.Volunteer_ID) == 0 &&
```

```
row7.ADDRESS_ID.compareTo(row5.Address_ID) == 0
```

\tLog Row Output – Rejected Data



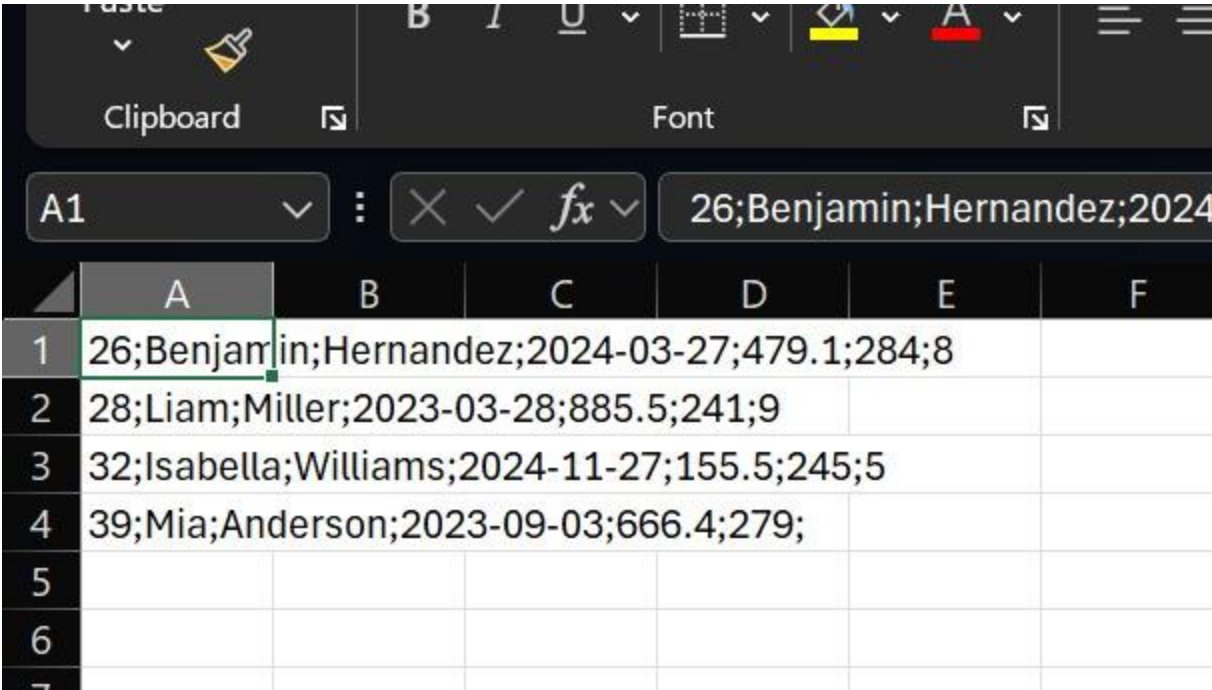
ORACLE OUTPUT:



The screenshot shows the Oracle Developer environment. The 'Connections' pane on the left lists the database schema 'prc-02'. The 'Query Result' pane displays the following data:

DON_ID	DONOR_FIRST_NAME	DONOR_LAST_NAME	DONATION_DATE	DONATION_AMOUNT	ADDRESS_ID	VOLUNTEER_ID
1	Elijah	Anderson	28-APR-23	26.9	102	4
2	Elijah	Williams	21-JUL-24	932.5	105	1
3	Harper	Johnson	22-APR-24	176.1	102	3
4	Benjamin	Davis	17-SEP-24	727.4	105	3
5	Charlotte	Smith	31-OCT-23	136	103	2
6	Elijah	Thomas	12-OCT-23	560.4	101	2
7	Elijah	Jackson	22-JUN-24	21.5	104	1
8	Lucas	Gonzalez	08-MAY-23	853.1	101	4
9	Logan	Taylor	26-OCT-23	424	101	1
10	Mia	Martinez	24-NOV-23	972	104	4
11	Sophia	Martin	01-JUN-24	855.9	102	3
12	Harper	Moore	18-NOV-24	780.7	101	4
13	Amelia	Martinez	20-OCT-24	734.7	104	2
14	Benjamin	Thomas	13-DEC-23	343.6	104	3
15	Lucas	Miller	18-MAY-24	161.4	103	2
16	Elijah	Smith	30-APR-23	948	101	1
17	Sophia	Martinez	16-JAN-24	673.7	101	4
18	Ava	Smith	15-JAN-23	244.8	103	3
19	Elijah	Taylor	15-FEB-23	882.9	104	1
20	Jacob	Smith	04-SEP-23	133.7	103	2

REJECTED CSV :



The screenshot shows an Excel spreadsheet with the following data in the 'A' column:

	A	B	C	D	E	F
1	26;Benjamin;Hernandez;2024-03-27;479.1;284;8					
2	28;Liam;Miller;2023-03-28;885.5;241;9					
3	32;Isabella;Williams;2024-11-27;155.5;245;5					
4	39;Mia;Anderson;2023-09-03;666.4;279;					
5						
6						
7						

CSV Creation through tFileOutputDelimited in Talend Job Design:

The screenshot displays the Talend Open Studio for Data Integration interface (version 8.0.1.20211109_1610) for a job named 'Job grkjb01'. The job design is as follows:

- Repository:** LOCAL W25-Group07-db6
- Job Design:**
 - Contexts:** Contexts
 - Code:** Code
 - SQL Templates:** SQL Templates
 - Metadata:** Metadata
 - Documentation:** Documentation
 - Recycle bin:** Recycle bin
- Job Design:**
 - Components:** tFileList_1, tFileInputDelimited_1, tMap_1, tFileOutputDelimited_1, tDBInput_1, tDBInput_2, tDBOutput_1
 - Flow:** The job starts with tFileList_1, which connects to tFileInputDelimited_1. tFileInputDelimited_1 connects to tMap_1. tMap_1 connects to tFileOutputDelimited_1. tFileOutputDelimited_1 connects to tDBOutput_1. tDBInput_1 and tDBInput_2 are also connected to tMap_1.
 - Statistics:** tFileList_1: 70088 rows in 17.33s; tFileInputDelimited_1: 4046.89 rows in 0.01s; tMap_1: 4 rows in 0.09s; tFileOutputDelimited_1: 16 rows in 0.07s; tDBOutput_1: 16 rows in 0.07s.
- Designer Code:**
 - tFileOutputDelimited_1:**
 - Basic settings:** Property Type: Built-in
 - Advanced settings:** Use Output Stream
 - Dynamic settings:** File Name: "C:/Sheridan College/Semester 2/DBAS32100 Database Management/Project/out.csv"
 - View:** Row Separator: "\n", Field Separator: ";"
 - Documentation:** Append, Include Header, Compress as zip file
 - Schema:** Built-in, Edit schema, Sync columns
- Palette:** Find component...
 - Recently Used:** tLogRow, tFileOutputDelimited, tMap, tFileInputDelimited, tFileList
 - Big Data:** Business Intelligence, Charts, DB SCD, Jasper, Splunk
 - Business:** Amazon, Azure Storage, Azure, Box, Dropbox, Google Drive, Google Storage, Google, Marketo, NetSuite, Salesforce, ServiceNow, Snowflake, iCloudStart, iCloudStop
 - Custom Code:** Data Quality, Databases, DotNET, ELT, Combined SQL, Connections